

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ



کابل پوهنتون



Microsoft Excel

استاد: پوهنیار عبداللہ صافی

What is a Formula?

A Formula is an expression used to perform calculations in Excel

- It allows users to compute results automatically

Formulas can include:

- Numbers
- Cell references
- Operators
- Every formula must start with an equal sign (=)

Real-Life Example

- Teacher calculates marks manually vs Excel
- Example:
Student marks $\rightarrow 20 + 30 + 25$
- **Explain:**
Excel does this instantly $\rightarrow$ saves time and avoids mistakes

Structure of a Formula

A formula includes:

- = (start)
- Values (numbers or cells)
- Operators (+, -, *, /)

Example:

=A1 + B1

Addition (+)

- Used to add numbers

- Example:

$$=10+5 \rightarrow 15$$

Explain:

Use addition for:

- Total marks
- Total salary
- Total expenses

Subtraction (-)

- Used to find difference
- Example:
 $=10-3 \rightarrow 7$

Explain:

Use in:

- Remaining balance
- Profit/Loss
- Age difference

Multiplication (*)

- Used for repeated addition
- Example:
 $=4*5 \rightarrow 20$

Explain:

Very important in:

- Salary calculation
- Price $\times$ Quantity

Division (/)

- Used to divide values

- Example:

= $20/4 \rightarrow 5$

Explain:

Used in:

- Average

- Ratio

- Per-person calculation

Four Basic Operators

Operator	Operation	Example	Result
+	Addition	=10+5	15
-	Subtraction	=10-5	5
*	Multiplication	=10*5	50
/	Division	=10/5	2

Important Note (Errors)

- Division by zero $\rightarrow$ ERROR
- Example:
 $= 10/0 \rightarrow \#DIV/0!$

Explain:

Students must avoid errors

Cell References

Why Cell References are Better

- Dynamic (auto update)
- Professional work
- Easy to edit

Explain:

Real Excel work always uses cell references

- Instead of numbers → use cells
- Example:

=A1+B1

Explain:

If A1 changes → result changes automatically

Cell Formatting

Excel allows us to format cells by changing:

- ❖ Background/fill
- ❖ Font
- ❖ Borders
- ❖ Alignment
- ❖ Number format
- ❖ Text orientation
- ❖ Formatting makes a spreadsheet more attractive and easier to understand

Applying Ready-Made Styles

- ❖ Excel provides built-in cell styles.
- ❖ Ready-made styles can be applied quickly.
- ❖ Styles can improve the appearance of a spreadsheet.
- ❖ Select a cell or range of cells.
- ❖ Open **Home** → **Cell Styles**.
- ❖ Choose a suitable style.

Example with Cells

- $A1 = 10$

- $B1 = 20$

- Formula:

$=A1+B1 \rightarrow 30$

Change $A1 \rightarrow 15 \rightarrow$ Result becomes 35

Simple Sum

- Add multiple cells

$=A1+A2+A3+A4$

Explain:

Good for small data

But not good for large data

SUM Function

Why SUM is Important

- Faster
- Less typing
- Less mistakes

Explain:

Always prefer SUM in real work

- Best method for total

=SUM(A1:A10)

Explain clearly:

A1:A10 means range from A1 to A10

Example of SUM

Student: Ahmad	Score
Math	80
English	75
Pashto	90
Computer	85

To calculate the total:

=SUM(B2:B5)

Result: 330

Why Formulas are Important

- Save time and effort
- Reduce human errors
- Perform complex calculations easily
- Automatically update results when data changes

What is a Function in Excel?

A **function** is a **predefined (built-in) formula** in Excel that performs a specific calculation.

Key Points:

- Already created by Excel
- Saves time
- Easy to use
- **Example:**

=SUM(A1:A10)

☞ This adds all values from A1 to A10

Difference Between Formula and Function

Formula	Function
Created by user	Predefined in Excel
Uses operators	Uses function names
Example: =A1+B1	Example: =SUM(A1:A10)

Creating a List

- ❖ A list contains related information.
- ❖ Lists can contain names, numbers, dates, or other data.
- ❖ Enter a heading at the top of each column.
- ❖ Enter each item in a separate row.
- ❖ Lists make information easier to read and manage.

Example:

1. Student Name
2. Age
3. Department
4. Score

Excel lists can be used to:

- ❖ Organize information.
- ❖ Sort data.
- ❖ Filter data.
- ❖ Calculate totals.
- ❖ Format information.
- ❖ Manage large amounts of data.